7th Grade Math · Weekly Check · Unit 10

C-AR.2.1

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities

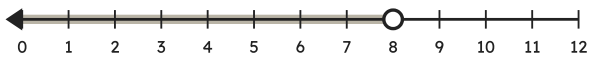
Student code: _____ Calculator: yes

Learning goal: I can write and solve one-step inequalities and graph the solutions. (circle your level after scoring)

4	I can write an inequality from a real-world situation, solve it, and interpret the solution.
3	I can solve a one-step inequality, including with negative numbers, and graph the solution.
2	I can graph an inequality on a number line.
1	I need help to start this benchmark.

1. The graph shows the solution of an inequality.

Which inequality has this solution?



A. $2.5x \leq 20$

B. $x + 2.5 < 20$

C. $2.5x > 20$

D. $2.5x < 20$

2. Solve the inequality.

$-3 + t \geq -6.9$

Turn over →

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities Student code: _____

3. Solve the inequality.

$-0.5n > 0.5$

Which number line shows the solution?

A. Open circle at –1, shaded left

B. Closed circle at –1, shaded left

C. Open circle at 1, shaded left

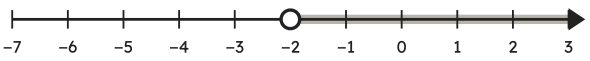
D. Open circle at –1, shaded right

4. The product of a number x and –0.5 is less than –5.

What are the possible values of x?

5. The number line shows the solution of one of the inequalities below.

Which inequality has this solution?



A. $x + 3.5 \geq 1.5$

B. $\frac{x}{4} < -0.5$

C. $x - 4.5 > -8.5$

D. $-1.5x < 3$

Score: ____ **/ 5** **My level:** **1** **2** **3** **4** **→ copy both to your progress strip**

Answer key — C-AR.2.1 · MA.7.AR.2.1 · Solve one-step inequalities

Unit 10: Equations and Inequalities. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **D.** $2.5x < 20$

The open circle at 8 shows that 8 is not included, and the shading to the left shows $x < 8$.

$2.5x < 20$ has this solution: dividing both sides by 2.5 gives $x < 8$.

$2.5x \leq 20$ would need a closed circle; $2.5x > 20$ shades the other direction; $x + 2.5 < 20$ has the boundary 17.5.

2. Regular $t \geq -3.9$

Add 3 to both sides.

$t \geq -6.9 + 3 = -3.9$.

3. Regular **A. Open circle at –1, shaded left**

Divide both sides by –0.5. Dividing by a negative number reverses the inequality symbol, so $n < -1$.

An open circle at –1 shows that –1 is not included, and the shading goes to the left.

4. Regular $x > 10$

"The product of a number x and –0.5" is $-0.5x$, and "less than –5" gives $-0.5x < -5$.

Divide both sides by –0.5. Because it is negative, the inequality symbol reverses: $x > 10$.

5. Rigor **D.** $-1.5x < 3$

The graph shows $x > -2$: an open circle at –2, shaded to the right.

Solve each choice. Only $-1.5x < 3$ gives $x > -2$ (dividing by the negative coefficient reverses the symbol).

The other three solve to a different direction, the other kind of circle, or a different boundary.

Re-check strip (3 items, C-AR.2.1-R)

1. Regular $x > 8$

"The product of a number x and –1.5" is $-1.5x$, and "less than –12" gives $-1.5x < -12$.

Divide both sides by –1.5. Because it is negative, the inequality symbol reverses: $x > 8$.

2. Regular **A. Closed circle at –3, shaded right**

Divide both sides by –1.5. Dividing by a negative number reverses the inequality symbol, so $c \geq -3$.

A closed circle at –3 shows that –3 is included, and the shading goes to the right.

3. Regular **$n \leq -5$; graph: closed circle at –5, shaded to the left. Score: 1 point for the solution (or an equivalent inequality) and a matching graph; either alone earns 0.**

Multiply both sides by –4. Multiplying by a negative number reverses the inequality symbol.

$n \leq 1.25 \times (-4) = -5$

Graph: –5 is included, so draw a closed circle at –5, and shade to the left because the solutions are less than or equal to –5.

Comparison strip (2 items, C-AR.2.1-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular $x \leq 8$

"The product of a number x and –2.5" is $-2.5x$, and "at least –20" gives $-2.5x \geq -20$.

Divide both sides by –2.5. Because it is negative, the inequality symbol reverses: $x \leq 8$.

2. Regular **$w \leq 5$; graph: closed circle at 5, shaded to the left. Score: 1 point for the solution (or an equivalent inequality) and a matching graph; either alone earns 0.**

Divide both sides by 0.5.

$w \leq 2.5 \div 0.5 = 5$

Graph: 5 is included, so draw a closed circle at 5, and shade to the left because the solutions are less than or equal to 5.

Reteach plan — C-AR.2.1

MA.7.AR.2.1 · Solve one-step inequalities

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities Student code: _____ Calculator: yes

1. The product of a number x and −1.5 is less than −12.

What are the possible values of x?

2. Solve the inequality.

$-1.5c \leq 4.5$

Which number line shows the solution?

A. Closed circle at −3, shaded right

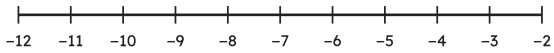
B. Closed circle at 3, shaded right

C. Open circle at −3, shaded right

D. Closed circle at −3, shaded left

3. Solve the inequality. Then graph the solution on the number line.

$\frac{n}{-4} \geq 1.25$



Score: ____ / 3 → copy to your progress strip

7th Grade Math · Comparison items (2)

C-AR.2.1-X

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities

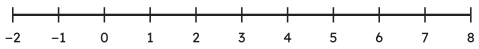
Student code: _____ Calculator: yes

1. The product of a number x and −2.5 is at least −20.

What are the possible values of x?

2. Solve the inequality. Then graph the solution on the number line.

$0.5w \leq 2.5$



Score: ____ **/ 2** → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-AR.2.1-X

Benchmark: **MA.7.AR.2.1** · Solve one-step inequalities

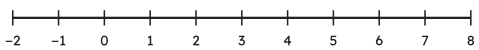
Student code: _____ Calculator: yes

1. The product of a number x and −2.5 is at least −20.

What are the possible values of x?

2. Solve the inequality. Then graph the solution on the number line.

$0.5w \leq 2.5$



Score: ____ **/ 2** → **copy to your progress strip**